RESEARCH PROPOSAL / SHORT PAPER OR POSTER

TopoPerturb: Robust Topological Responses to Genetic Perturbations

A reproducible analysis of cellular-state reorganization in single-cell transcriptomics using CausalBench K562 and RPE1.

CENTRAL PROBLEM

Do gene knockdowns induce reproducible changes in the topology of single-cell expression-state distributions, and do these changes provide information beyond standard measures of perturbation magnitude?

01 Motivation and scientific gap

Genetic perturbation studies are commonly summarized using differential expression, shifts in average expression, and distances between cell populations. These approaches do not explicitly describe multiscale connectivity or loop structure in the expression-state landscape.

Persistent homology offers quantitative topological summaries, but high-dimensional noise can distort them. Damrich et al. (NeurIPS 2024) demonstrated that graph-derived spectral distances can improve topology detection in noisy single-cell data. Our proposed study extends that setting to **intervention-versus-control comparisons** - a question not answered by topology detection alone.

Research questions

RQ1	Which perturbations exhibit topological changes beyond matched control-sampling variability?
RQ2	Can topology distinguish perturbations with similar conventional expression effect sizes?
RQ3	Are topological responses consistent or context-specific across K562 and RPE1?

Proposed contribution

- **Primary:** A null-calibrated, perturbation-specific analysis of persistent homology in single-cell data.
- **Empirical:** A controlled comparison of topological response with differential expression and distributional baselines.
- **Interpretability:** Mapper views of selected robust changes, followed by cross-cell-line testing.

Novelty status: This is a hypothesis and proposed framework - not an established novel algorithm or a demonstrated biological finding.

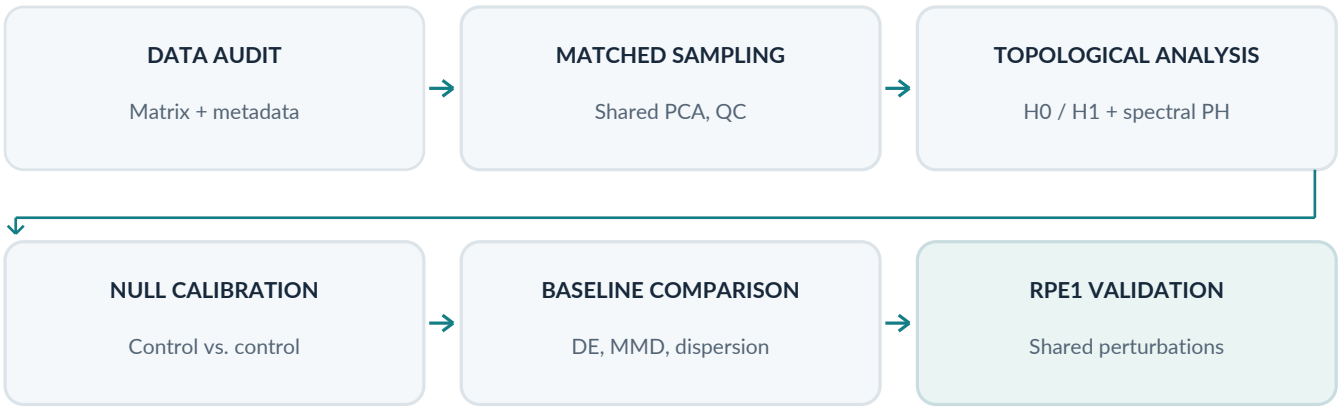
02 Data and experimental workflow

Data sources

Dataset	Role	Source
K562 - leukemia cell line	Method development and pilot	Hugging Face: causalbench-k562
RPE1 - retinal epithelial cell line	Independent cell-context evaluation	Hugging Face: causalbench-rpe1

Before analysis: Verify mirror contents: cell-by-gene matrix, expression normalization, control labels, perturbation-target names, guide IDs, batch labels and cell counts. Published CausalBench statistics are not a substitute for inspecting these mirror files.

End-to-end workflow



Methodological stages

- **1. Pilot selection.** Select 5-10 K562 perturbations with adequate cells and diverse conventional effect sizes. Exclude poorly represented or confounded groups.
- **2. Shared representation.** Apply consistent QC, appropriate normalization, variable-gene selection and one common PCA fit within each cell line (start with 20-50 PCs; sensitivity test).
- **3. Quantitative topology.** Compute 0D (connected components) and 1D (loops) Vietoris-Rips persistence using Euclidean and kNN spectral distances. Use matched subsamples and standardized parameter settings.
- **4. Qualitative Mapper.** Create Mapper graphs for a small number of representative perturbations, labelled by perturbation and available cell metadata. Treat branches as structural patterns, not temporal trajectories.

Comparability safeguard: Compare diagrams only after checking distance-scale comparability. In spectral analyses, inspect the effect of group-specific graphs versus a shared reference graph; harmonized parameters alone may not produce comparable scales.

03 Quantification, baselines and controls

Primary outcome: topology-response effect

Let D_0 and D_g denote the persistence diagrams for a control subsample and a gene-g knockdown subsample, constructed in a comparable metric space. For each independent matched draw, compute a diagram distance such as a p-Wasserstein distance:

PERTURBATION-LEVEL EFFECT

$d(g) = W_p(D_0, D_g)$ | Compare repeated d(g) values against distances from independent control-versus-control draws.

Summarize a perturbation using its median matched-sample distance, empirical significance versus a control null, and a bootstrap uncertainty interval. An optional standardized contrast can be defined as (median[d(g)] - median[d(null)]) / (MAD[d(null)] + epsilon), but only if its numerical stability and interpretation are checked.

Baselines: what must topology add?

Comparator	Question it answers
Differential expression	How broad or strong is the transcriptional response?
Centroid shift	Does the average cellular expression state move?
Energy distance / MMD	How different are the overall distributions?
Within-group dispersion	Is the perturbation associated with changing heterogeneity?
Euclidean vs. spectral PH	Are topological results robust to high-dimensional noise and metric choice?

Controls and robustness tests

- **Balanced sampling:** Match cell counts across groups, repeat draws, and report confidence intervals.
- **Negative controls:** Control-versus-control draws, label-shuffling tests where exchangeability is justified, and perturbations with weak conventional effects.
- **Technical effects:** Examine guide identity, batch and library depth; avoid treating many cells from one experimental batch as independent biological replicates.
- **Sensitivity:** Test PCA dimensions, kNN sizes, distance normalization, outlier handling and number of sampled cells; control multiple testing across perturbations.

Decision criterion for a credible paper

The most compelling result would be two well-sampled perturbations with similar conventional expression-effect magnitudes but **distinct, statistically reproducible topological signatures**. Correlation or significance alone is insufficient: results should survive the controls above. A null or negative finding remains informative and should be reported accurately.

04 Paper outputs and references

Figures and tables for the final submission

- **Figure 1:** PCA/UMAP plus selected Mapper networks, with clearly described labels and controls.
- **Figure 2:** Persistence diagrams and null-calibrated perturbation responses with uncertainty.
- **Figure 3:** Topological score versus conventional distribution-effect measures; highlighted matched-effect perturbations.
- **Table 1:** Dataset sizes, perturbation coverage and sample inclusion. **Table 2:** Quantitative comparison and robustness summary.

Suggested 2-4-page paper outline

Abstract (motivation, question, method, principal verified results); **1 Introduction** (gap and contribution); **2 Data & Methods** (controls, metrics, null calibration); **3 Experiments & Results** (baselines, selected perturbations, sensitivity); **4 Discussion** (biological interpretation and limits); **References.**

Boundaries and interpretation

No causal edge direction is inferred from topology alone. The datasets are cross-sectional: Mapper branches cannot be labelled trajectories without independent evidence. Cell-line-specific responses are not automatically generalizable to patients. Computationally expensive spectral and H1 analyses should use carefully designed subsampling.

Key resources

1. K562 mirror: huggingface.co/datasets/huggingbahl21/causalbench-k562

2. RPE1 mirror: huggingface.co/datasets/huggingbahl21/causalbench-rpe1

3. Damrich, S., Berens, P. & Kobak, D. (2024). *Persistent Homology for High-dimensional Data Based on Spectral Methods.* NeurIPS 2024.

4. CausalBench: Chevalley et al. (2022 preprint; later benchmark publication). *A Large-scale Benchmark for Network Inference from Single-cell Perturbation Data.*

Project status: Proposed design only. Dataset schemas, selected perturbations, statistical feasibility, biological significance, and novelty still require verification.